



DEPARTMENT OF ECONOMICS
JAHANGIRNAGAR UNIVERSITY
TUTORIAL ANSWER SCRIPT

Marks

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Tutorial No. 08

Course Title: Labor Economics

Signature of the invigilator:

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ANSWER FROM HERE

Ans. to the Ques NO-2(a)

The general on-the-job training refers to those that raises productivity for all firms in the labor market, not only the firm that provides the training.

The specific on-the-job training refers to that which raises the productivity for only the firm that provides the training, the skill is not transferable.

Without training, an individual generates MRP₀, given that the training does not depreciates the productivity over time. It gives the productivity pattern EFG.

With training it generates the ABCD schedule.

During training the productivity falls as one less time in production and MRP falls to MRP_1 .
 training it rises to MRP_2 .

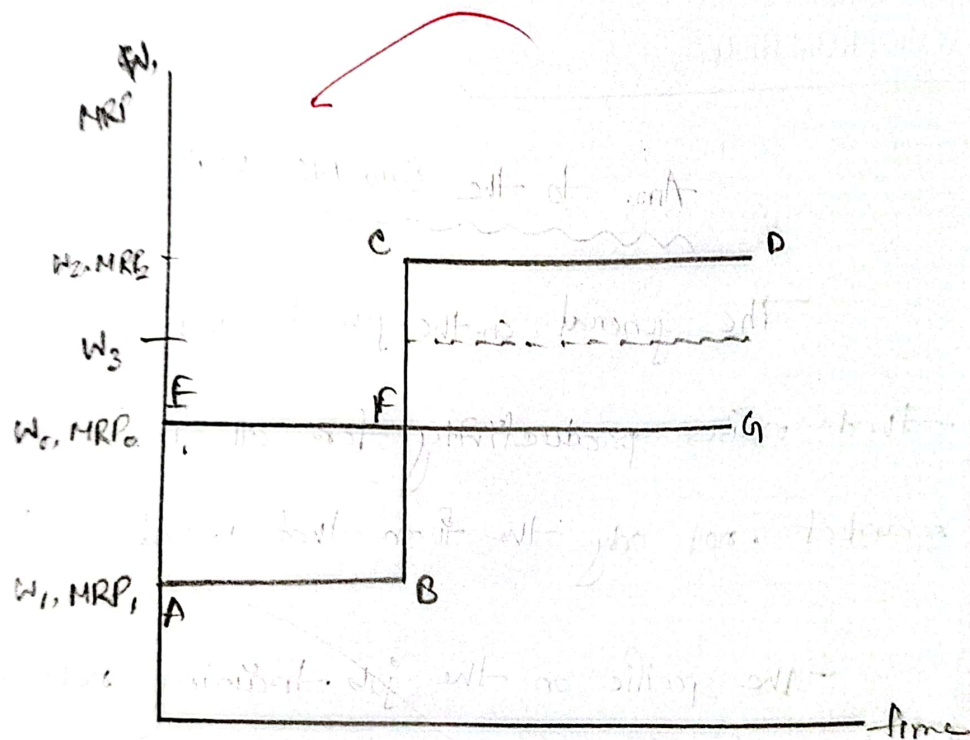


fig: OTJ productivity pattern

the cost of training is given by area AEFB.

the benefit of training is given by area FEDH.

In the case of general OTJ, the worker bears

the cost of training by receiving lower wage W_1 .

during training period. Because in a competitive market, the wage is W_0 , then with transferable general OTJ training period, the firm has to pay W_0 , even when the MRP is MRP_1 , and it hopes to gain by paying W_0 with MRP_2 . But the competition eventually raises the wage up to W_2 when workers have MRP_2 .

So the workers bear the cost by receiving W_1 during training and gains by obtaining W_2 after the training ends.

In case of specific OTJ training, the firm pays the cost of training. It provides W_0 wage even when MRP_1 . As the training raises MRP to MRP_2 only for that firm, but with non-transferable skill, the workers' MRP remains MRP_0 in all other firms so the market wage is at W_0 . Then as

the training ends, the firm continues to pay even with MRP_2 . It gets the return on the investment in this way.

In reality the firm shares portion of their gain say $(W_3 - W_0)$ by paying a higher wage, say W_3 , than W_0 . This happens as workers and firms creates Bilateral monopoly and both have bargaining powers.

Ans. to the Ques NO-2(b)

The major implications of on-the-job training model that validates the theory are —

① Employee turnover:

The specific ~~of~~ OTJ training reduces the employee turnover by quitting and layoffs.

If the employee quits (or forced to quit) the job, then the firm loses all the investment on the training. And the employee loses by getting lower wages elsewhere ~~by~~ due to the non-transferable skill.

② Minimum wage law:

It reduces the amount of general OTJ provided. Usually the young people don't remain on the same job for a longer period, so they have to pay for the general training by accepting

lowers wage during the training period, say the wage of w_1 . But if minimum wage floor was set to w_0 where $w_0 > w_1$, then the firm has to bear the cost, which it doesn't want. So the firms curtail general OTJ training for young employees.

③ Internal Labor Market:

The specific OTJ training is a key factor of creating internal labor market. If all skills could be acquired by general training, then the firms can hire workers from the external labor market.

④ Area of Indeterminacy:

In perfect competition, there is law of one wage. But with specific OTJ training, the workers become heterogeneous, and it creates an area of indeterminacy in wage rate.

⑤ Age/Earnings Profile:

Assuming only the formal education as general training and with no further training and no depreciation the age/earnings profile is ~~ABEHF~~ ^{ABEHF}. But in reality we get the profile ~~ABEHF~~ ^{ABEDG} with parabolic Dn portion where earnings is increasing at first, then falls after middle age.

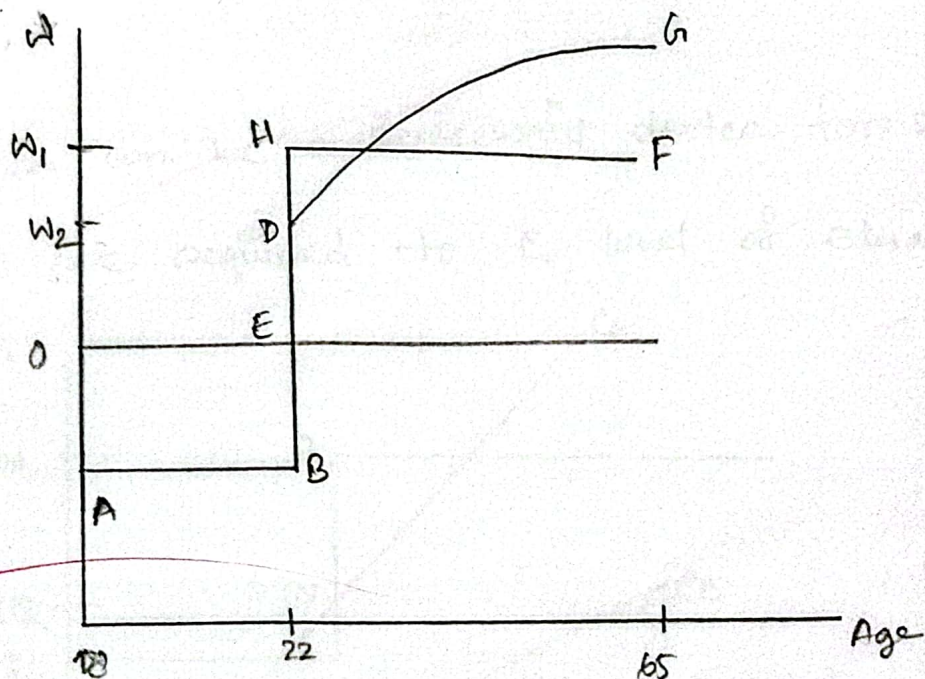


fig: age learning profile

At first one pays for the general training, so receive lower wage W_2 than W_1 . Then with specific

As saving continues to rise and λ starts to fall due to human capital depreciation

the economy converges to a steady state where the capital stock is constant

and the growth rate is zero

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Ans to the Ques NO-3(a)

In the employee selection process let's consider two groups of workers.

A: Low ability workers with $MRP_L = \$10$

B: High ability workers with $MRP_H = \$20$.

Without any successful ~~screening~~ screening device both get the average wage, not equal to their MRP_L .

Education can be a successful device for selection.

Assume all are required to E_0 level of education.

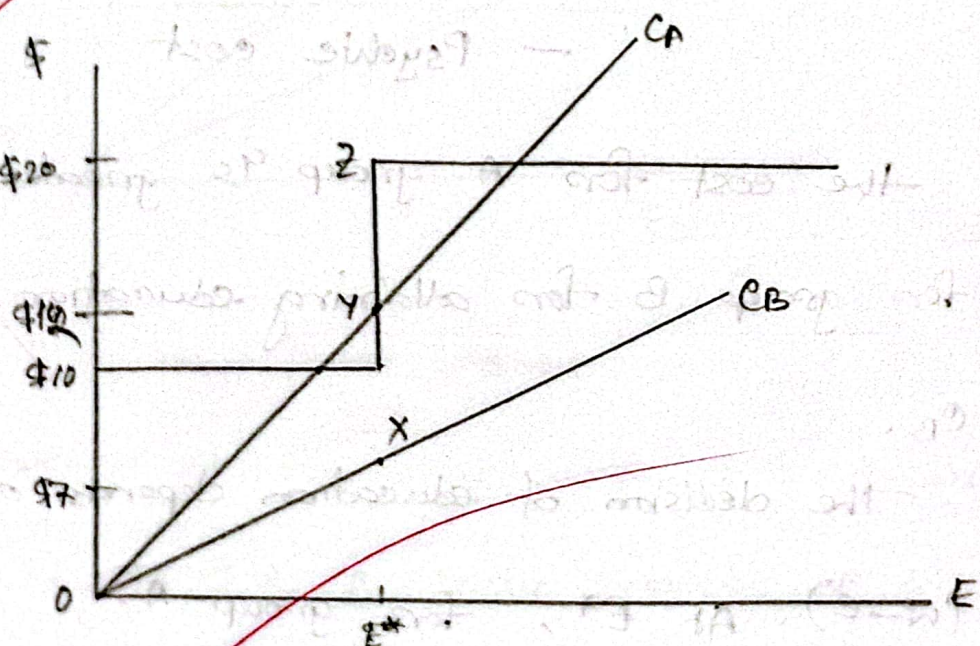


fig: education - selecting device.

Based on past experience, the firm decides that below E^* level, everyone is low ability and paid \$10 and above E^* everyone is of high ability and paid \$20.

The success of E^* depends on the cost of education. Assuming attaining E^0 level requires zero cost, the cost of education is negatively related to productivity.

So due to — Direct cost

— Indirect/opportunity cost

and

— Psychic cost

The cost for A group is greater than the cost for group B for attaining education. C_A lies above C_B .

The decision of education depends on net wage ($w - c$). At E^* , for group A,
Net wage is $= (20 - 12) = \$8$.

at E_0 it is \$10.

for group B, at E^* , net wage \$13 and at E_0 it is \$10. So A group receives E_0 and B gets E^* level of education, and it separates two group successfully.

Any point other than E^* is not successful.

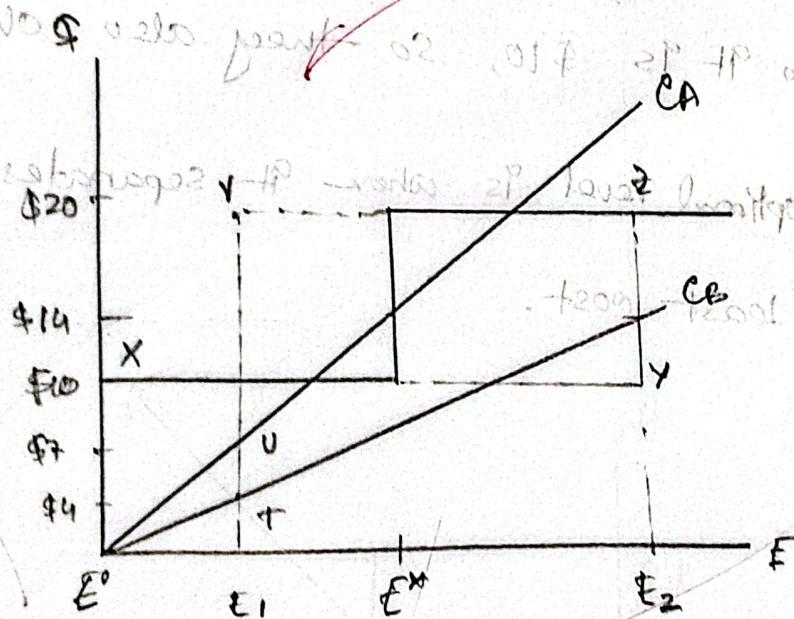


Fig: signal failure.

for A group, at E_1 , net wage \$13 greater than at E_0 of \$10. A group chooses E_1 level.

for B group, at E_1 , the net wage $\$20 - \$4 = \$16$

greater than that of E_0 of $\$10$. So they also

choose E_1 . It doesn't ~~separate~~ ^{separate} groups.

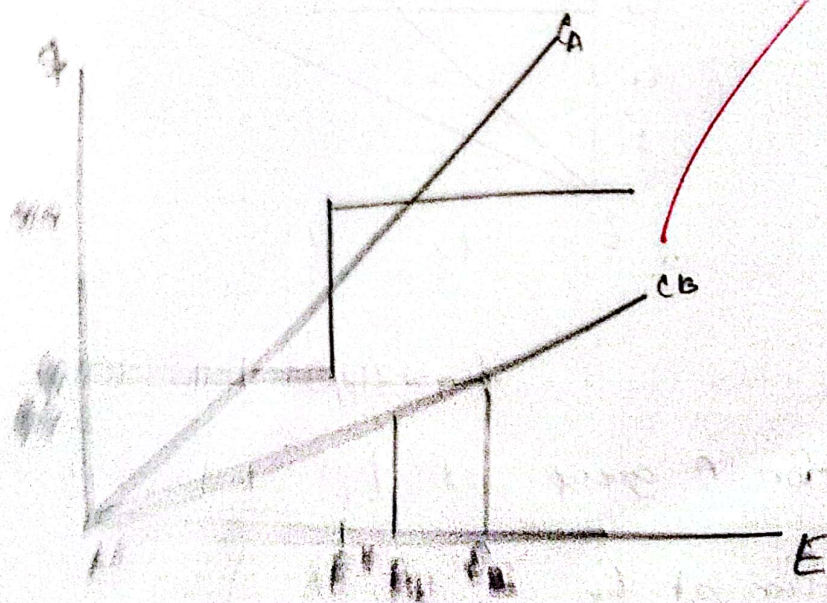
for A group,

At E_2 , the net wage is negative as $C_A > W$. So chooses E_0 .

for group B, at E_2 , the net wage is $\$6$,

at E_0 it is $\$10$, so they also choose E_0 level.

The optimal level is when it separates two groups with least cost.



The optimal E

for B group, at E_1 the net wage $\$20 - \$4 = \$16$

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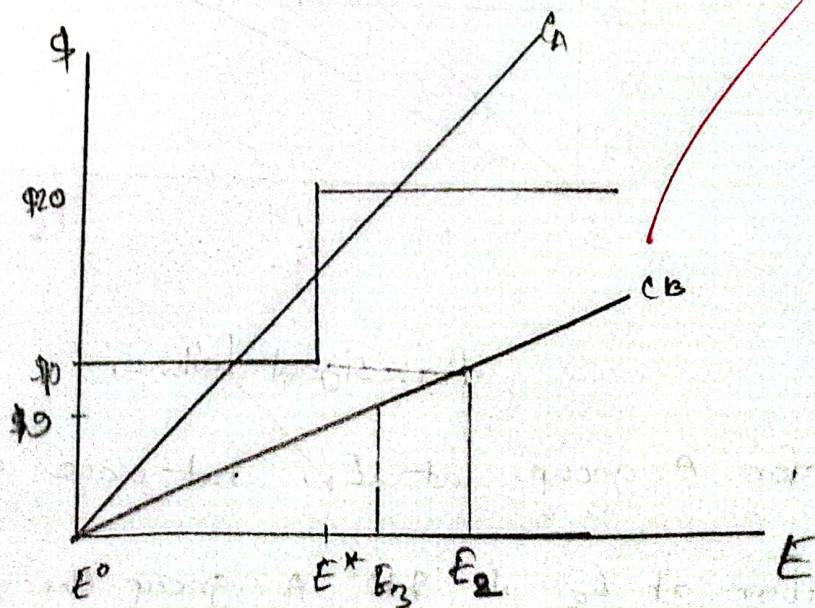
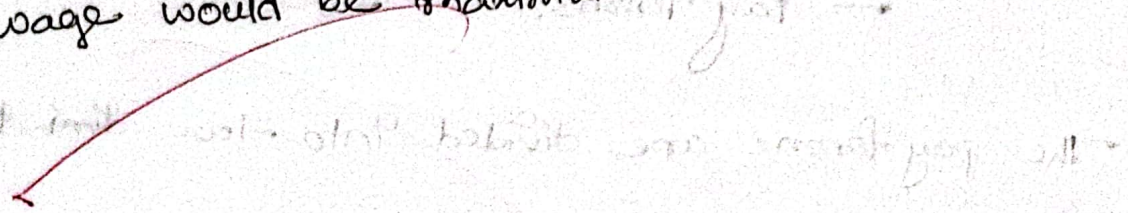


fig: optimal E

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the optimal level is somewhere between E^* and E_2 where the net wage is maximum, like a level E_3 with net wage \$11. So the optimal level is closest to E^* where the net wage would be maximum.



(1) to (2) (the size of 1)

(2) to (3) (the size of 2)

these two types of off-diagonal elements are not symmetric. The difference between the two types of off-diagonal elements is that the first type is the difference between the two types of off-diagonal elements and the second type is the difference between the two types of off-diagonal elements.

Ans to the Ques NO - 3(b)

In the compensation management, there are two aspects

- Pay level

- Pay forms.

→ The pay forms are divided into two different dimensions

- Size based

- Performance based.

→ The two options of size based pay forms are

(a) Individual based (the size of 1)

(b) Group based (the size of N).

→ There are some advantages and disadvantages of

these two types of pay forms. We discuss those

respect here —

(a) Individual based pay forms :

In this category there are wage/hour, salary/year (input-based) also sales rate, commission rate (output based).

The advantages of individual based pay forms are that they motivate self-improvement. As one gets the payment based on his or her skill, it motivates him to do better to get a higher income. This works as the best incentive to motivate the workers to do their job in the best possible way.

But the drawback is that it discourages teamwork. So if any production process has severe interdependencies or complementarities that requires teamwork, the individual based pay forms is not that effective.

(b) Group based pay forms:

This is the opposite of individual based. Some examples are ~~contracted~~ profit-sharing, gain-sharing.

The advantage of this pay form is that it encourages group work and production process with much interdependent conditions, required this form for efficiency.

The problem with this is the free riders problem. As one gets paid for team-work, they want to avoid doing their job properly. It is known

as $\frac{1}{N}$ problem. As the size of N rises, it becomes less observable to free ride. So the problem rises.

In this case quarterly gain sharing among workers seems to be more effective than company wise profit sharing.